

The University after AI

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Abstract

This paper develops an analytic framework to offer strategic guidance to rebuild or reform universities through the AI era. We review the history of universities, the history of technology in universities, the economics of AI, and the economics of the university in order to combine these analytic perspectives into a framework to identify the main impact pathways of disruption of the university after AI. Universities have been impacted by technology before, with little effect. This time is different. Earlier waves lowered the cost of content, but content is not and has never been the core business of the university, which is matching and verification. Universities are multi-sided markets whose strategic asset is a credible warrant of quality over a credence good and AI is the first technology to disrupt this core function. We propose a range of strategic implications, both defensive and offensive, to deal with this disruption.

Executive Summary

The university as an institution is a thousand years old and technology has threatened before without consequence. Books, radios and computers all lowered the cost of producing and distributing educational content, with the university mostly unchanged. This was because content was never what universities sold. They sell matching and verification: bringing students, scholars, employers and funders together, and warranting to third parties the knowledge claimed. AI is the first technology to reach both functions and so the first with the capacity to disrupt the university's business model.

We develop an economic framework for what follows and what to do about it. It treats the university not as a cloister or a factory but as a multi-sided market (a platform) coordinating at least fifteen distinct groups, sustained by cross-subsidies between them, whose strategic asset is a credible warrant of quality over a good whose quality nobody can directly observe. Four mechanisms follow. First, signal collapse: because AI lowers the cost of producing assessable work by roughly the same amount for able and less able students alike, the credential stops separating them, and it does so even when no rule is broken. Second, disintermediation: AI substitutes for the informational sides of the platform, which are also the paying sides, and it disintermediates the institution from its scholars as well as its students. Third, relocation of the strategic asset: administration was valuable as the mechanism of quality assurance, and AI automates the mechanism without supplying the function, so the asset moves from operating coordination systems to being a credible verifier. Fourth, a shift in where rents accrue, from stocks of transmitted knowledge toward the capacity to adapt to new technologies, and from competent execution toward exploration. Three things are untouched: co-production, the peer cohort, and anything requiring physical presence or legal standing.

We set out ten strategic proposals, grouped by what the university must own, what it should build, and what it should reshape. The central proposals are to separate certification from instruction institutionally; to relocate the costly signal into modalities AI cannot enter; to occupy the training gap left by AI adopting firms; to build fast accreditation capacity; to treat the cohort as a designed product; to host scholars rather than employ them; and to specialize in the exploration that private incentives now undersupply. Because a multi-sided platform cannot change one side without renegotiating all of them, the practical route is greenfield. The university has survived a thousand years mostly by founding new colleges, not by rebuilding old ones.

1. The life cycle of a university

A university is a self-governing micro-republic, corporate in form, approximately 1000 years old (older than nation states, younger than religion), that exists within but always apart from society (for instance they often have their own police and justice tribunals), and which serves as the custodian of knowledge, creator of new knowledge, and whose primary business is to upload that knowledge into a select group of the next generation. This is a service that students want and that society is willing to pay for.

Our story begins in 11th century Paris and Oxford, where students would gather to collectively hire and negotiate fees with learned scholars, who would instruct and examine them (Compayre 1893). This scholarly collective would rent the resources they needed, including room and board. The professors would certify the learning. The university originated as a focal point (Schelling 1960), where students knew to find academics and where academics knew to find students. Universities began as a place and a platform, an institutional solution to demand for warranted knowledge. That fundamental structural logic hasn't changed in a thousand years.

Universities developed, in the early centuries, to train certain classes of society: religious scholars and jurists in the first instance, philosophers and classic humanities scholars next, medicine subsequently, and scientists, engineers and business schools much later and surprisingly recently. The research university was invented in 19th century Germany and imported to America in the 20th. Around the world, universities massively expanded their remit to society after the Second World War, moving from an experience for the elite to mass participation. As more countries become wealthy (e.g. Asia, Middle East) they invest more in universities, which in turn have grown and improved. Universities have ever sought to be 'universal' in their approach to knowledge, rather than specialized. In most developed economies, it is normal for a substantial proportion (as high as 40 percent) of high-school students to proceed to university. As this has happened, universities became more interwoven with the state, and the state often recognized the advantages of this (Glaeser et al. 2007). Most universities today (but not all) are considerably reliant on public financing, directly through grants and subsidies or indirectly through student support or tax advantages. Consequently, they are less independent, more political, and more complex to govern than their earlier forms. A most striking pattern over the past 50 years is the growth of university administration.

Universities today, viewed as a distribution, are simultaneously wealthy, thriving, more important than ever, and going through an existential crisis of loss of social trust, institutional bloat and decay in organizational form and incontinence in costs, and deep shock and disruption to its business model due to the cumulative erosion of a recent series

of powerful digital wave trains. The first wave began in the 1980s with computers and the internet, through to social media which delivered a digital society. Universities adapted to that: students, for instance, were among the first to get email. The second wave came with the rebuilding of a digital economy (platforms, cloud, crypto, cyber, IoT, etc.) which shifted the knowledge students needed. Universities adapted to that: students, for instance, were offered new courses in cloud computing and blockchain. The third wave, where we are now, is AI and agents. This rebuilds intelligence and how it fits into society and the economy. This paper is about how universities might adapt to that, building on previous work with colleagues (Davidson et al 2020, Allen et al 2022, 2025).

But new technology has happened before. In the 15th century, the printing press arrived and shocked the world, driving a dramatic fall in the cost of knowledge and accelerating its distribution through society. Incredibly, while this technology fractured great religions and powered the emergence of nation states and vast corporate empires, it somehow didn't much effect universities, who adapted by building great libraries. Six centuries later, the digital information revolution is transforming universities. It is a curious fact that much of the technologies of the digital revolution came from universities, as public research, they have been among the least affected by it compared to other parts of society (especially market economies). Where many other industries have been profoundly transformed by digital technologies (media, finance, retail, transport, manufacturing, etc.) universities still look, in their basic shape and activities, remarkably like how they did 50, 100 or even 200 years ago. The ancient covenant that places universities somewhat outside society and the economy, as self-governing republics, seems to also place them outside technology.

Now previous waves of information technology (books, salons, radio, television, photocopiers, computers, internet, digital media, search engines, social media, MOOCs, etc., Montgomery et al. 2018) have also threatened the university over the centuries, and nothing much came of it. Why is the latest wave of AI, since about 2018, different? The answer is that earlier technology waves lowered costs of making and distributing educational content, but content was never the product that universities sold. It didn't matter that it went to free. Rather universities sell matching and verification. By lowering the cost of matching (and administration) and the costs of producing the outputs by which knowledge has conventionally been verified (research, papers, coursework, etc.), artificial intelligence is the first wave to actually reach these factors, and so the first information technology to disrupt the business model of the university in hundreds of years.

2. AI in universities

Today, as AI washes across the university, a large literature has bloomed studying its effect. Main topics are the effects on administrative automation, teaching and learning productivity, and research productivity, as well as dissociation with traditional labor market alignment and loss of credential value. But academics being what they are, scholarship has mostly proceeded field-by-field.

In education studies, much worry about assessment, both validity and security. Lots of small sample surveys and reporting of problems. In cognitive learning science, much worry about cognitive offloading (i.e. how it relates to learning and how it undercuts the importance of desirable difficulty). The scholars of learning, on the whole, don't like AI much, at the most benign seeing it as a 'pedagogical inconvenience' (Gilman 2026). Nor do the philosophers and critical theorists, who fret about whether we will ever return to the true humanistic university, as a pedagogical force to shape character and virtue through dialogue. Political economists and sociologists (e.g. STS scholars) worry ever more about concentration of power in the proprietary stack of technology, and of ownership and control of data. 'Bright satanic mills', they wittily call them. Literature journals are shot through these days with autoethnographies of dysfunctional current states and dystopic speculative fiction of possible futures of the university.

The mostly negative valence of the humanities and critical social sciences contrasts with AI-impact literature in the more STEM fields, which have emphasized the acceleration and falling costs of automation of research (Wang et al. 2023, Messeri and Crockett 2024, Hao et al. 2026). This is mostly seen as improving the productivity of science and advancing human knowledge and is therefore good for universities and, downstream, for economies and society. As of September 2026, mathematicians have now mostly made their peace with the fact that the latest models are better at theorem-proving than they are, and that their job going forward is mostly going to be verification of the model's work and guiding the models toward interesting problems. As goes math, so goes the rest of science.

Business schools have been more rigorous about specifically and statistically what is happening and how it might unfold, a program led by economists by building and testing models of how AI will affect markets, firms and economic growth. There is a large body of literature here, but the two canonical models are (1) that AI is a reduction in the cost of *prediction* (Agarwal et al 2019), which is a species of *automation* (Acemoglu and Restrepo 2019), which can be studied through the dynamics of *tasks* (Acemoglu and Autor 2011) and (2) a macro endogenous growth model that predicts a 'slow explosion' (Jones and Tonetti 2026) where automation happens quickly in some tasks but is bottlenecked (also called 'weak links') in tasks that require judgment and verification, because most jobs are

bundles of tasks that AI cannot always unbundle (Catalini et al. 2026). The effect of AI automation of tasks on labor markets is therefore complex, with some jobs being undercut by AI and de-skilled, and others being supported by AI (Brynjolfsson et al. 2025, Autor and Thompson 2025, Garicano et al. 2026, Jovanovic and Rousseau 2026). AI is automating some jobs out of existence, but it is also, as a countervailing force, creating new work (Autor et al. 2026) and accelerating learning in firms (Beraja and Talamas 2026). There is growing evidence that the wage premium, which has long existed for college educated workers and underpins the economic logic of private investment in university education, has begun to rapidly decline since 2022 (Azar et al. 2026), which overlaps with the AI era, but there is as yet no causal identification. Hassan et al. (2026) isolate a growing skill premium that accrues during times of rapid technological change, where the excess rents to young people working in new fields and new jobs accrue to their usefulness in dealing with new technologies. It follows that universities may be especially valuable during an AI revolution, if and only if they can prepare students for that AI revolution, a theme we return to in section 6 below.

Imbued with the task model, and with an entrepreneurial spirit, business school researchers have directed these insights back at the university by focusing on the unbundling thesis of disruptive innovation (e.g. Gilman 2024, 2026), which reverses Clark Kerr's famous thesis of the rise of the modern 'multiversity', as 'a social infrastructure of knowledge transmission'. Just as labor economists are seeing AI unbundling jobs into component tasks, some of which can be automated away and delivered by AI at near zero marginal cost, the same logic applies to universities, which are historically bundled tasks of research, teaching, credentialing and so on, some of which can be effectively unbundled and delivered at new zero marginal cost.

The AI and universities disruption literature overlaps with the general findings of new digital technologies on jobs, careers and organizations, which emphasize complementarity. As Google economist Hal Varian explains: "If you are looking for a career where your services will be in high demand, you should find something where you provide a scarce, complementary service to something that is getting ubiquitous and cheap." The same logic explains which firms profit from technological innovation (Teece 1986, 2018), that, applied to universities, requires that new business models will need to identify and deliver complementary services to an AI economy take-off.

Despite all the grumbling, even as many academics are trying to time their run to retirement before the singularity breaks over the university, nobody seriously now argues the university survives AI unchanged. Yet each field has only diagnosed the part of the university it can see — assessment, learning, ownership, research productivity, etc — and

what is missing is a theory of the organization that holds them together. Economics is useful here because it models how falling costs or changes in capabilities in one activity propagate through the bundles that hold activities together. That apparatus has been mostly focused so far on firms and labor markets, but it can also be usefully applied to universities. To take aim, we need to first establish the basic economic logic of a university.

3. Simple economics of universities

The community of scholars inside any university is often deeply romantic in their thinking. The philosophy of the republic is that the university stands apart as a community of learning and whose value lies in the cultivation of mind. This vision shapes how they understand their employment, their obligations, and the public standing of the institution. But it offers little strategic guidance to leadership beyond soliciting unconditional public funding in an era when governments are increasingly reluctant to provide it.

Governments and employers, and to a certain degree parents, in turn, are much more utilitarian. Theirs is an industrial model of the university, as a factory that produces credentialled graduates along with valuable byproduct of commercially and socially useful research. This vision does speak to funders, but it doesn't explain why teaching and research are bundled, why the institution's main revenue line is treated by its own staff as an annoyance, or why the administrative apparatus appears so organizationally bloated.

Yet the university is neither a cloister nor a factory. It is better understood as a platform (a multi-sided market). To see this, consider the product being produced and consumed. Education is a complex product. It is an *information good* (Arrow 1962), an *experience good* (Nelson 1970) and a *credence good* (Darby and Karni 1973). The production description of lectures, textbooks, contact hours and so on can be specified, but it carries little signal. The real product is an experience good for students and a credence good for their employers, as what they've actually hired, or the value of the certification, will not become clear for years, if at all. It is also a *private good*, as the student directly benefits from education, but with *public good* properties, as there are positive externalities to an educated population, as well as to research (Nelson 1959). Universities are also *local public goods* with powerful regional effects through agglomeration and knowledge spillovers (Valero and Van Reenen 2019). Universities are open knowledge institutions (Montgomery et al. 2018) and education is also *co-production* (Ostrom 1996), as cooperation of both student and teacher are jointly required to produce the good. Higher education is a private investment good subject to severe finance constraints, since human capital cannot be collateralized, so public financing is efficiently applied on the demand

side. Higher education is also a *third-party good* (with *derived demand*), as students choose programs in anticipation of employer preferences. Furthermore, there are two competing theories about the nature of the good produced. *Human capital theory* (Becker 1964) holds that what is produced is a type of capital, embodied in a human (hence human-capital), as an increase in productive capacity that is monotonic with investment of time and resources in education. *Costly signalling theory* (Spence 1973), on the other hand, argues that the value of a higher education credential is a reliable signal of an unobservable quality that was, crucially, already present before education, but unobservable. The purpose of the education, and why it is long and expensive, is to separate honest signals of quality from fakes. In this view higher education is privately valuable but socially wasteful (Caplan 2018). Different actors in education are also producing different things. Students are also purchasing access to a social network, but they themselves are also inputs into that network for other students (and thus value depends on exclusion as much as inclusion). Same for professors, researchers, donors and alumni. All these factors make the demand side of higher education extremely complex.

Note that Spence's separating equilibrium requires the cost of producing the signal is monotonically decreasing in the unobservable type. The high-type student finds the degree relatively less costly to obtain than the low-type student, which is what makes the credential informative about qualities that would otherwise be unobservable. AI has two effects here. It makes rule-breaking easier and cheaper, which raises enforcement costs. But even when these costs are incurred and new rules are followed, it also compresses the difference between what high-type and low-type students must spend to produce the same graded output. This shifts the market toward a pooling equilibrium in which the credential is weak at separating types, and willingness to pay for it falls.

On the supply side, universities are a labor-intensive service sector with limited scope for productivity growth, with wage costs set by productivity elsewhere in the economy (Archibald and Feldman 2011). Universities raise all the money they can and spend all they raise, so costs follow available revenue not production requirements. Technology has been the standing candidate for escaping the cost disease yet has repeatedly underdelivered. Market structure is neither competitive nor monopolistic. Universities operate under monopolistic competition with strong reputational differentiation, high entry barriers created by accreditation and regulation, and prestige hierarchies that are remarkably stable over time, generating durable rents at the top and thin margins below (Hoxby 2009). Governance and capital structure are correspondingly unusual. Universities are non-profit or public entities with trustee boards rather than shareholders, tax-advantaged endowments, and internal constituencies with quasi-formal participation rights

(Hansmann 1990, Aghion et al. 2010). For governments, universities are regional and national infrastructure. For philanthropists, universities are a vehicle for reputation and mission returns. For private capital, universities are an asset whose value lies in brand equity and tax treatment rather than in cash flows.

A university is a strange economic object. The product it produces is not what it seems. It is not education *per se*, sold by the unit, but credible claims about an unobservable quality that attaches to a person. The production process of this warrant is complex, involving luring resources in through a nexus of overlapping promises and mutual consensus. The evolved governance of this strange multisided institutional creature is built for maximizing trust in the production of a good no one can see by a process few can understand.

And now disruption, from a wave of powerful new information and governance technologies, bringing what Allen et al. (2025) call ‘institutional acceleration’. But disruption in a multisided market (Allen et al. 2022) is disruption of coordination, not production. And so disruption of universities will not look like Christensen's (1997) standard model of industrial disruption, with a cheaper entrant taking the low end of a market. A platform is attacked through its cross-subsidies, so the damage appears on one side, and the failure appears on another. Digital technologies change not how we organize production but how we organize exchange. AI disrupts universities because coordination and verification, across a complex network of internal exchange among stakeholders, are what the university has been selling all along.

4. A university is a platform

A university, then, is neither a cloister nor a factory but a *platform*, or a multi-sided market whose primary economic function is to reduce the transaction costs of matching distinct groups who wish to trade with one another. This view is developed by Allen et al. (2022), which originally sought to explain the effect of the joint shocks of Covid and digital technology on the university. The argument connects two literatures: *transaction cost economics* to explain organizational form by the structure of coordination costs (Alchian and Demsetz 1972, Williamson 1985, Hart 1995) and the *economics of multi-sided markets* (Rochet and Tirole 2003) in which the firm coordinates network externalities and supplies governance (Evans and Schmalensee 2016). Allen et al. (2022) argue that universities are among the most complex multi-sided platforms in the economy, identifying at least fifteen distinct sides, as stakeholders: undergraduate students, international students, research students, teaching faculty, research faculty, industry as employers, industry as research customers, professional associations, federal government, state and local government, alumni, donors, high schools, parents, and local business and suppliers.

Each side derives value from the presence and quality of the others. Undergraduates benefit from strong research students, who make good tutors; research students benefit from strong undergraduates, who create demand for their labor, and from strong research faculty, who make good supervisors. Employers value access to a screened talent pool, governments value graduates, national innovation capacity and regional development, and alumni value the ongoing quality of the platform as it determines the value of their own credential. They also, in the US, where philanthropic alumni are especially active, value sports. The production function of a university is to create value by orchestrating these matches.

Universities have been platforms since the first cohort of students in Paris and Oxford sought to match with scholars in the 11th century. The late 20th century expansion of universities has just added more sides to the platform as more stakeholders were engaged (regional development, alumni and philanthropy, human capital policy, industrial policy, innovation and technology policy, business, etc.), and with increasingly heterogeneous and often conflicting objectives, universities became ever more complex and difficult to manage and strategically direct. Unlike even enormous industrial corporations or banks, universities do not have dominant principals (e.g. shareholders) nor single objectives (e.g. profit maximization). The task of senior leadership is the continuous negotiation of side-payments and cross-subsidies that keep the market clearing. Many otherwise puzzling features of university finance become legible on this reading: the cross-subsidy from international to domestic students, from high-margin to low-margin disciplines, and from teaching revenue to research activity, which in turn generates the reputation that attracts the students who fund it. This account also explains the growth in administrative scale, not as evidence of bureaucratic capture or inefficiency, but due to the coordination burden a fifteen-sided market imposes.

The strategic architecture of the university as a platform follows from its competitive position, and defining three types of university (elite, specialized, and normal), Allen et al. (2022) locate the key strategic asset the university must own (or, not allow anyone else to own), which they call a ‘Hart asset’¹ (Davidson et al. 2020), and which identifies what assets a university must strategically own to avoid opportunistic exploitation and hold-up problems, etc., in a competitive environment. For elite universities, the Hart asset is brand and reputation. For specialized universities, the Hart asset is expertise, but for most normal universities, the Hart asset is administration. This is a difficult concept to grasp because it

¹ The concept of a ‘Hart asset’ was developed by Davidson et al. (2020) and is named after Oliver Hart, who theorized the logic of ownership in a firm as a solution to problems of opportunism under uncertainty (as first operationalized by Oliver Williamson, building on Ronald Coase’s transaction cost logic). “A ‘Hart asset’ is an asset that a firm cannot strategically afford a rival firm to own or control due to the risk of hold up, and therefore must be held within the firm, and upon which a profitable business model can be built.”

runs utterly counter to the Professoriate's self-perception, as being the most important part of the university. But the economic logic is that the academic staff are the most visible part of the university, not the most strategically important. The Hart asset of the modern university is the smooth and flawless operation of its administrative protocols.

There are two structural consequences. First, because the university as a platform is a complex nexus of cross-subsidies, no side or component can be evaluated in isolation. The loss of a side invariably withdraws the transfer that was sustaining participation elsewhere. Second, the Hart asset is defined functionally rather than physically. Administration is the strategic asset of the modal university because it is the mechanism through which quality assurance is delivered under conditions where quality is unobservable. Any technology that separates the mechanism from the function will therefore relocate the asset rather than simply automate it.

5. Mechanisms by which AI impacts the university

We now ask how AI, which brings a massive fall in the cost of applied intelligence, affects a fifteen-sided platform whose strategic asset is a warrant of quality over a credence good, and whose organizational logic lies in minimizing the transaction costs of making matches and verifying outcomes. Any theory of the firm must answer four questions: (1) what the organization sells, (2) what it exists to economize, (3) what it must own, and (4) where its rents come from. Prior to AI the university's answers were: (1) a warrant over a credence good; (2) the transaction costs of a fifteen-sided market; (3) the mechanism of quality assurance; and (4) reputational differentiation under monopolistic competition. AI changes all four, which we set out as the four mechanisms below. We then consider what the technology does not touch, and how fast an institution of this form can move.

5.1 Signal collapse (what the university sells)

The credential as a separating equilibrium is informative because the cost of obtaining it falls with the unobservable quality it certifies. AI acts on both terms of that relationship but at different speeds. It is quickly and already lowering the cost of producing the observable outputs by which knowledge is assessed (essays, problem sets, code, analyses, theses). On a longer horizon, it is more slowly lowering the cost of acquiring the underlying capability those outputs were supposed to indicate. The empirical work on AI in education finds a consistent dissociation between performance and learning (students using AI complete tasks faster, score higher and retain less). Cheating has become easier, but the actual strategic problem is that the cost of the signal is increasingly independent of type.

When the same output is available to the able and the less able at approximately the same low cost, the monotonicity condition fails, separation fails, and the equilibrium moves toward pooling (Spence 1973). The credential becomes uninformative at the margin, which is a different and worse problem, because the institution's revenue depends on the credential while its social licence depends on the information.

The private return to the degree is expected to fall and to fall before any change in the underlying human capital because the return is, in part, a return to the signal. The recent decline in the graduate wage premium (Azar, Gine and Sanz-Espin 2026) is consistent with this. We should also expect that employers will substitute toward direct screening, including independent tests (where legal!), work trials, portfolios, task-based assessment, and increasingly their own AI-assisted evaluation of candidates. That substitution disintermediates the university's certification function and it is already observable in hiring practice. Plainly, universities can be expected to resist employers seeking to substitute their own independent verification of quality instruments.

5.2 Disintermediation (what the university economizes)

A multi-sided platform exists because matching is costly (Rochet and Tirole 2003, Evans and Schmalensee 2016). AI lowers matching costs but not evenly, and sides whose relationship with the university is purely informational are exposed. Specifically, undergraduate teaching materials, literature search and synthesis, tutoring, matching industry research to expertise, and much of the advisory and pastoral functions that operates through information transfer are all activities in which a general-purpose model is today already a good partial substitute. These models and agents, including harnesses, are likely to keep getting better on these frontiers.

The sides that hold and become binding constraints on automation are those where the university's role depends on something other than information: legal standing and regulatory recognition (professional accreditation, licensure, visa sponsorship), physical co-presence (laboratories, clinical training, residential formation), long-lived trust that must be attached to a durable entity (alumni networks, donors, government research funding), and warrant of quality to third parties.

Disintermediation also runs in the other direction, through the supply of scholarship rather than the demand for it. Allen et al. (2022) argued that faculty are not Hart assets because there is no hold-up problem: a productive academic who leaves takes her funding, students and reputation with her, and the institution has no residual claim on capital it never owned. Artificial intelligence sharpens this considerably. Where research once

required institutional infrastructures of support (such as RA, library and database access, IT, etc.) the latest generation models and agents now substitute for almost all of it. (Travel grants for conference attendance are mostly untouched!) The minimum efficient scale of a research operation is falling, in some fields toward a single person. The academic therefore becomes less dependent on the institution exactly when the university's revenue is under pressure from the demand side. The university is being disintermediated from the students who pay and from researchers who produce at the same time.

Strategy here is tricky. Face-to-face instruction need not disappear but note the sides most easily disintermediated (instruction, which can be substituted with AI tutors, for instance) are, in most universities, the sides that pay directly. Undergraduate and international student fees are the transfer that funds research, which generates the reputation that attracts the students. Allen et al (2022) described how these cross-subsidies clear a 15-sided market. Disintermediation attacks it from the revenue end and signal collapse attacks it from the product end. A platform that loses a paying side loses the capacity to sustain the participation of sides that were never paying their own way. The immediate risk to research from artificial intelligence is not that AI does the research. It is that AI removes the undergraduate teaching revenue that was buying the research in the first place.

5.3 Hart asset relocation (*what the university must own*)

The Hart asset of the modal university (neither elite nor specialist) is administration (Allen et al. 2022). A growth strategy for such institutions is to expand by supplying the administrative platform on which others (other units) transact. But AI makes administrative coordination among the most exposed activities in the institution. But administration was never the strategic asset *per se*. Rather, it was identified as the core strategic asset to protect against opportunism because it is the mechanism through which the university performs quality assurance over a credence good and quality assurance was the activity that could not be safely delegated to a counterparty. The mechanism of quality assurance was properly the Hart asset. The arrival of AI enables the administrative mechanism to be substantially automated but that alone, as a productivity gain, does not supply the underlying function of quality assurance, which is bundled with the universities processes and reputational capital, amongst other factors. A model for instance can schedule an exam, mark a script, issue a record, etc, but it cannot warrant to a third party that the person holding the record possesses the capability, because the warrant requires an entity independent of the production of the thing being warranted, that bears reputational cost for error, and that persists long enough for that cost to matter. The Hart asset under AI therefore relocates from operating the coordination systems to *being a credible verifier*.

(This transformation from operation to verification is a point also made by Catalini et al 2026 in the context of the evolution of jobs after AI.)

A university that outsources its administrative stack while retaining its examining and certifying authority has given away a commodity and kept the asset. A university that automates its examining while retaining its administration has done the reverse and has no defensible position. Note also that this is complementary with signal collapse, which destroys the informational value of the conventional credential and, by the same movement, creates a market for institutions that can restore it. Scarcity of credible verification is produced by the abundance of cheap output.

5.4 Where the rents move (how the university can profit)

The skill premium is a function of the age structure of technologies in use (Hassan, Kalyani and Restrepo 2026). Skilled workers hold a comparative advantage in learning new technologies, an advantage that decays as those technologies standardise. The returns to higher education are therefore predicted to rise with the rate of technology creation, not with the stock of knowledge transmitted. A period of rapid technology creation is the best possible environment for higher education if the product being sold is adaptation rather than content. The product graduates need is not a body of knowledge to fit a career, but a demonstrated capacity to acquire the next technology faster than the market can standardise it. But universities, and accreditation cycles, still work as if access to expert-quality knowledge is the scarce resource.

Research is being disrupted by AI too. The cost function for research has fallen on multiple margins (data, collaboration, analysis, writing and reviewing) and so research productivity has sharply accelerated (Hao et al. 2026). Structurally, AI-augmented research appears to automate established fields, rather than open new ones.

AI is affecting what is scarce, and therefore where value is created. Under AI exploration of new technologies in the labour market and of new questions in science is scarce while competent execution within established domains becomes abundant and commodified. This is especially notable as some of these domains were previously high in expertise and difficulty, e.g. literature reviews, statistical analysis, theorem proving. That predicts where rents will accrue that a university after AI can capture.

AI makes the conventional academic output cheap to produce, which destroys the informativeness of the credential and creates a market for credible verification. It lowers matching costs asymmetrically, disintermediating the informational sides of the platform,

which are the paying sides, breaking the cross-subsidy on which everything else depended. And it raises the return to adaptation while lowering the private return to exploration, opening a gap that a public-purpose institution is well-placed to occupy.

5.5 Invariants

Several invariant factors survive disintermediation. These are worth noting because they also identify targets for relocating the costly signal. The first is *co-production*. Education requires the joint effort of teacher and student (Ostrom 1996), and no technology that lowers the cost of producing outputs supplies the student's willingness to be changed by the process. The second is the *peer effect*. Students are outputs of education, and buyers of the product, but they are also inputs into the good other students consume, so value depends on exclusion as much as on inclusion. AI does nothing to this. It cannot manufacture a cohort, and the assortative matching that produces a valuable peer network is if anything more valuable when other forms of differentiation are collapsing. The third is *embodiment* and *legal standing*. Laboratories, clinical placement, residential formation, licensure, accreditation and visa sponsorship all require a durable, physically located, legally recognized entity, and are not substitutable by a model.

6. Strategic proposals

"For everything to remain the same, everything must change" (Tancredi Falconeri, in *Il Gattopardo*, 1958)

What strategic implications follow? We consider 10 proposals, grouped by the question they answer (what must the university own? what should it build? what should it reshape?). The overarching constraint is that because a university is a multi-sided platform, no single side can be altered unilaterally. This is why incumbent reform efforts, especially as adaptive responses to new technology, tend to stall at pilot stage. Universities are hard to strategically direct not because of intransigence of coddled academics, or really for any cultural reason, but rather because they are massively multi-sided platforms that are resistant to renegotiation and brittle where changes are forced. Disruptive innovation theory (Christensen 1997) reaches the same conclusion from the perspective of market analysis, namely that the incumbent cannot pursue the disruptive model because doing so cannibalizes the business that currently pays. The proposals below are therefore easier to implement in new institutions or campuses, or in ring-fenced units, than in the core of an existing university. It also implies a structural advantage for systems currently building new capacity rather than defending old capacity.

Against this stands the view, probably the majority position today, that AI is a productivity opportunity to be absorbed into the existing bundle, that teaching, research, examining

and credentialling along with the secondary services of libraries, journals, pastoral care and the rest can each be better delivered with AI, all while the institution retains its shape. The central message of this report is that this is unlikely. Universities will need to change to survive.

6.1 What the university must own

We start with the core assets and positions the university must own, building on the model of Hart assets presented above (Davidson et al. 2020, Allen et al. 2022).

Proposal 1. Institutionally separate certification from instruction

The Hart asset of the university after AI is the warrant, not the teaching. The strategic form this implies is a university that examines and certifies as its primary business, and that treats instruction as a service which may be produced internally, purchased, or left to the student. Almost no one working inside a university right now wants this. But it is the logical direction this new wave of technologies is pushing the university. And so the first universities to get there and successfully implement this strategy will capture the high and valuable ground.

In concrete terms this means open entry, examination-only pathways to a degree, and pricing on the assessment rather than on tuition or contact hours. There are precedents for this model, for instance in the University of London External system, in professional bodies such as the CFA, ACCA, in actuarial and medical boards, and in competency-based providers, which together demonstrate that the model is legally and commercially viable at scale. These are, admittedly, essentially accreditation agencies and very different types of business from the modern university. The proposal is that the modern university should become more like them.

A verification monopoly, however, is only valuable if it is credible, and it is only credible if it is independent of the parties whose output it verifies. This is the same design problem as central bank independence: separate governance, separate funding, tenure protections for examiners, and a formal prohibition on the certifying body having a financial interest in pass rates. A university that certifies its own teaching has always been conflicted, which was previously resolved as a long-run repeated game (staking reputation, which proxies university quality by age) or with the support of national and international certifying boards and agencies (TEQSA in Australia, NCAAA in Saudi Arabia, AACSB for business schools, and so on). A significant fall in the cost of generating fake signals that at the same time lowers the cost of content delivery, as the same technology does both, shifts the

equilibrium toward verification and raises the strategic value of in-house verification assets.

Separation raises an immediate question about who holds the academic protections. The protections must attach to the warrant, not to the teaching. Tenure exists so that an examiner can fail a candidate, reject a result, or refuse a certification without personal consequence. It is therefore the certifying body that must inherit the collegiate form (i.e. peer judgement, protected tenure, academic freedom) while the instructional arm can be run on ordinary managerial lines. This inverts the usual assumption that academic protections belong to the teaching faculty and administrative discipline to the back office. (See also Proposal 10.)

Proposal 2. Restore the costly signal

Signalling theory predicts that when a signal becomes cheap to produce, the equilibrium signal will adapt or evolve so as to relocate where cost is higher. The current pathway for many universities has been to increase investment in detection of cheating and increase punishments for those caught. This has the advantage of preserving investments in existing instruction operations, and so minimizes disruption costs, and mostly has buy-in from academic staff, administrators and sellers of third-party software. Yet it is already abundantly clear that it is an arms race, and worse, one that pits the university against its customers in a way that it is unlikely to win without significant concessions to fairness and equality. The assessment literature has already largely conceded the point that this is a losing strategy. And yet we persist.

A better approach is to move the assessment to a modality where the production cost remains type-dependent. At this stage, the best candidate is probably a return to the viva, common in earlier times as an observed, embodied, high-bandwidth, unscripted examination conducted as an oral defense, or as live problem-solving before an examiner. Variations might include extended practical demonstration or structured disputation, but, whatever the specific form, the model is a return to a bespoke real-time production model of quality verification. This is expensive per student and is complementary with cheap AI-delivered instruction. The basic strategy is to capture the reduction in instruction cost as an internal resource and relay it to making examination costly again.

The deeper point concerns where human judgment retains scarcity value. Catalini et al. (2026) argue that as execution becomes cheap, rents migrate to verification, which requires accumulated tacit knowledge, including judgment and taste. As Jack Clark, co-founder of Anthropic puts it:

"We're going to need to be extremely intentional about working out where we as people specialize so that we have that intuition and taste – or else you're just going to be surrounded by super-productive AI systems, and when they ask you what to do next, you probably won't have a great idea."

Ironically, perhaps, the pedagogy this implies looks like the traditional elite liberal arts education, or the Oxbridge tutorial system, or the classic MBA case study approach, as a simulation engine for building taste, judgment, refinement and intuition. The form of instruction that mass higher education abandoned as unaffordable might well return as ancient technology for building a future-proof product. If Catalini et al. (2026:1) are right, that:

“the binding constraint on growth is no longer intelligence. It is *human verification bandwidth*: the scarce capacity to validate outcomes, audit behavior, and underwrite meaning and responsibility when execution is abundant.”

Then it follows that the production technology for that same human verification bandwidth is going to be proportionately expensive. There is a lovely historical symmetry here, in that the collapse of the written signal returns us to the viva and the medieval disputation, which is what examination was before cheap paper made written assessment scalable. Note also that signal collapse and productivity gain are the same event. AI makes cheap precisely those effortful tasks whose costliness generated the signal, so a university cannot capture the full productivity gain and retain the handicap. It must choose, and if it wants to retain a costly signal it will need to migrate that signal somewhere newly costly. Perhaps riding well-bred horses and dueling might make a comeback.

6.2 What the university should build

Beyond the rebalancing of instruction and verification, the next strategic play for the university after AI is to look for new opportunities to rebuild value added. Above, we saw that value is moving to the training that firms are no longer providing, to adaptation, and to invariants AI does not touch. These are opportunities.

Proposal 3. Rebuild the missing junior loop

The university business model is, on average, that students pay for their general training in fees, and then firms hire them and train them further in specific skills until they become high value productive workers. Newly hired graduates often have low value to the firm because they still lack value-producing skills, and so early in a career the firm effectively pays wages that exceed marginal product. Firms nevertheless find this a good trade to make, rather than charging graduates to work for them, or paying much lower wages (as in

the older apprenticeship model), because firms have monopsony power arising from asymmetric information about worker quality that lets them appropriate part of the rent from training (Mincer 1974, Acemoglu and Pischke 1998).

Artificial intelligence erodes this bargain on both sides. First, it removes the junior tasks through which learning occurred. Second, it improves external screening, which reduces the informational rent that made training profitable. The reduced incentive of firms to train workers in industry-specific or even firm-specific skills is the 'missing junior loop' identified by Catalini et al. (2026), and it is a market failure with an obvious institutional remedy. Universities could exploit this opportunity by extending the training they do into this skill-specific space in the form of, for instance, brokered and supervised post-degree placement, structured apprenticeship years, or simulation-based training for judgment in domains where the junior rungs have been automated away.

The difficulty in this proposal is not specification and content delivery, which universities already know how to do, and which is often embedded in professional body accredited courses, for instance. Rather, the new opportunity is to design an effective financing mechanism to create and appropriate the rent of asymmetric information about revealed quality of each student. In the old model, firms invested privately in training because they could appropriate the return through asymmetric information and monopsony power in the labor market with respect to recently trained graduates. Universities cannot appropriate through that channel, but they can appropriate through the durable relationship with the graduate and with the cohort. We return to opportunities now possible with digital finance in Proposal 9 below.

Proposal 4. Fast-response accreditation

If returns accrue to adaptation to new technologies (cryptocurrency, drones, AI agents) rather than to acquisition of, and competence in, a stock of existing knowledge (finance, accounting, wet-lab biology), then the binding constraint is the speed at which an institution can stand up a credible qualification in an emerging technology. Content is not the constraint, as content is now almost free. The binding constraint in a world of rapid technological change is accreditation authority together with the availability of institutional expertise and examiners who can verify quality in a new domain.

The proposal is a standing unit with delegated authority to create, certify and retire short qualifications on a timescale of weeks, funded from the adaptation rents from new technology (Hassan et al. 2026). Universities are the only bodies that hold both the certifying authority and the disciplinary expertise required to make such qualifications

credible. They currently deploy that combination on a multi-year committee cycle, which is to hold a scarce asset and then guarantee it arrives after the rent has dissipated. The strategic and pecuniary value of this capability is directly proportional to the rate of technological change. Through an AI-driven hard economic acceleration (Jones and Tonetti 2026, Beraja and Talamas 2026) this may be a critical university dynamic capability.

Proposal 5. Sell the cohort

AI cannot manufacture a cohort, and the peer network is a good whose value depends on exclusion as much as on inclusion. As other forms of differentiation collapse, quality assortative-matching appreciates. Indeed, it is the reason elite universities are largely insulated from all four mechanisms, and it is available, in more modest form, to less-elite institutions on whatever margin of assortative-matching they can create and find value in.

The proposal is to treat cohort composition as a designed product rather than as a byproduct of admissions flows. This means selecting for complementarity, rather than only for rank, and building the institutional forms in which a cohort actually becomes a network, such as colleges, cohorts that progress together, structured collaboration, and residential and co-curricular density. This will require pricing and marketing the network explicitly rather than folding it into tuition. This asset will be extremely hard to build, as it requires deep curation and extreme selectivity (Y-Combinator, a Palo Alto-based technology incubator, is an example of this mechanism at work, by cohort). The challenge is to figure out how to build this at scale without diluting the brand. The tension with existing revenue models is direct and unavoidable. An institution that has been selling access at volume cannot also sell scarcity of access.

Proposal 6. Host scholars rather than employ them

The primary disruption AI brings is signal collapse and dislocation of the Hart asset. But a second disruption AI brings is to disintermediate the university from its researchers and from its students at the same time. As models substitute for research assistance, database access, computation and much routine scholarly labor, the minimum efficient scale of a research operation falls, in some fields toward a single person. The employment relation was underwritten by the scholar's dependence on institutional infrastructure. That dependence is weakening. At the limit it breaks entirely. What the university can still supply is legal standing to hold grants and employ people, ethics and compliance authority, physical facilities where these are required, co-presence and colleagues, and above all the warrant. A self-published work by an unaffiliated researcher (including an AI

agent, for instance) faces exactly the credence problem the university exists to solve. Affiliation is a certification of a researcher in the same way a degree credential is a certification of a student.

The proposal is therefore to unbundle affiliation from employment, making the university more like a club (see also Potts et al. 2017, on journals as clubs) or as a species of commons (see Potts 2019 on innovation commons) or what De Filippi and Santolini (2022) call an *extitution*. Rather than the binary of tenured staff or nothing, universities can offer graded affiliation over institutional standing, grant-holding capacity, facility access, and the right to publish under institutional imprimatur, priced as a service and subject to quality review. This converts a fixed cost into a revenue line, retains scholars the institution could not otherwise afford to employ, and extends the warrant into the market for independent research.

An extension of this proposal is to fundamentally redesign the employment contract and remuneration structure of a university, and which would only be feasible in a large greenfield university. The legacy university has most academics on comprehensive long-term employment contracts. This bundles compensation for high-risk difficult to monitor activities (frontier research) with low-risk easily monitored activity (teaching) with professional duty and governance of the republic obligations (committee work, conferences). It is a risk pooling model, where the university carries the residual risk and gates entry, capturing upside reputation on successful bets. Because of reputation pooling, there are strong incentives for mutual peer monitoring, especially at entry. But an alternative model is now viable with digital data oracle and finance technology using a design Potts and Rennie (2024) call a *contribution system*. The model would separate out piece-work components, which can be easily monitored and have medium-level substitutability (teaching, research supervision, some committee work) and tender these roles to competitive contract. Academics could choose as much or little as they please, potentially contracting to multiple universities. This would form one revenue stream for each academic and would be a cost line-item to the university. The second part are contribution systems that form over multiple teams across the university (a research group or center, a school or cluster, etc.) that pools all outputs and records contributions to the pool (the mechanism and logic is set out in Rennie and Potts 2025). Individual academics become, in effect, equity partners in these pools, which can be liquidated by share on demand (the university itself acts as a market-maker) or held as wealth. Funders contribute not to universities central accounts but directly to these contribution system pools based on the expected value of the work. The difference to the legacy model is that academics are incentivized to create economically and socially valuable work in voluntary

communities (Potts 2019) and then derive a mixed income stream that endogenizes their own risk tolerance and effort.

Proposal 7. Move to the new frontier

Hao et al. (2026) document a private-social wedge in research in which adopting AI raises individual output and career outcomes while contracting the collective range of enquiry. Wherever such a wedge exists, the activity on the losing side is undersupplied by private incentive and is then the natural business of a non-profit, public-purpose institution.

The proposal is that universities deliberately specialize in the exploration end of the distribution, low-data domains, unfashionable questions, methods that do not reduce to prediction, long-horizon work and negative results, and that funders re-weight toward it. This is what Hollis Robbins (2026) has called, in an essay on this theme, the unknown-unknowns (after the Rumsfeld matrix). AI has commoditized knowledge and particularly its search and discovery, including in domains that were previously high in expertise and difficulty such as literature review, statistical analysis and theorem proving. The implication is that the comparative advantage of universities has moved further afield, into the harder to conceptualize and reach corners of new ideas and possibility. This is simply to realize that the comparative advantage of universities is always on the *knowledge frontier* and that the frontier is always moving out, and so universities must ever seek to move out in front of it. That will always be hard to do, but do it they ever must.²

6.3 What it should reshape

Proposal 8. Shed sides

The common response to shrinking revenue in modern universities is to find efficiencies in operations, or reduce funding on some margins. The platform model of the university (Allen et al. 2022) says this is not in general survivable and that the alternative to involuntary contraction is chosen or strategic contraction. The platform-based strategy is more radical. It requires dropping one or more sides of the platform altogether then rearranging all trades among the remaining sides to reflect that loss of stakeholder demand and supply. The strategic exercise is to identify which sides the institution can serve better than any available substitute, and to exit the others through sale, spin-out, or transfer to a shared provider. The logic we have set out above suggest to retain sides whose

² This is not the same as the *resource frontier*, which also moves out with new knowledge, but this is the domain of entrepreneurship and innovation (see Beraja and Yuchtman 2026).

participation depends on legal standing, physical co-presence, durable trust, or warrant. In turn, exit sides whose participation depends on information transfer alone.

The growing literature on the effect of AI in the economy emphasizes a race between automation (the substitution of AI capital for labor over tasks, called the *displacement effect*) and the creation of new jobs (the *reinstatement effect*, Acemoglu and Restrepo 2019, Acemoglu et al. 2026). The net effect of AI on jobs and the economy depends on which effect dominates (and on the elasticity of substitution across tasks). A similar argument can be made about the effect of AI on the university but seen through the effect on platform architecture of the organization rather than task architecture of a job. Shedding sides are the displacement effect of AI automation on the university. The new activities discussed above are the reinstatement effect. The net shape of the university depends on which effect dominates.³

Proposal 9. Make the claim on graduate outcomes explicit

A university is already a residual claimant on the outcomes of its graduates, as their success determines its reputation, which determines its capacity to recruit and to raise funds. That claim is implicit, unpriced, and only realized through hopeful expectation (i.e. alumni donation). If the credential's signal value is falling, the institution has an interest in converting it into an explicit contract. Instruments might include outcome-contingent claims on the earnings differential, income share agreements (e.g. personal equity contracts), cohort-level contribution systems, internal prediction markets on programme outcomes. An alternative structure is an option on the graduate pool, in which the institution holds a right to make an offer, like a draft round in professional sports. Note that governments have an even stronger durable channel through the tax system, which is why HECS-type schemes of public loans repaid through the tax system have come to dominate higher education funding in many parts of the world. The proposal here is that universities acquire a version of the same instrument at institutional scale.⁴ The obvious challenges are adverse selection, enforcement of options contracts, and liquidity of markets that need to be created. The track record of income-share agreements is poor and the political economy of a university holding financial claims on its own graduates is difficult. These are

³ Note the platform case has a parameter with no counterpart in the task model (tasks are substitutes in production, so the elasticity of substitution governs how far labor reallocates). But in a platform sides are not substitutes but are complements bound by cross-subsidy. What matters is not how easily one side replaces another (elasticity of substitution) but how much of the transfer structure collapses when a side is withdrawn. A platform with loose cross-subsidies can shed sides and reorganize, whereas a platform with tight cross-subsidies cannot shed anything without renegotiating everything.

⁴ A certifying body must not hold a financial interest in the outcomes it certifies, so an outcome-contingent claim on graduates cannot sit with the entity that examines them. The claim belongs to the instructional and training arm.

in principle soluble with good mechanism design, but this is certainly a more speculative proposal. The argument in favor, however, is that it would become more like an overlapping-generations perpetual society or financial cooperative (like a building society) and is structurally similar to the contributions systems pooled-equity model discussed above in proposal 6, but here is built around students rather than faculty.

Proposal 10. Governance as the production technology of the warrant

Universities are difficult and complex to manage because they are everywhere a massively multi-sided platform business. In consequence, the standard outside business-consultant diagnosis of the problem – that rococo medieval governance produced cloistered and bloated organizations rife with underperformance, compounded by weakly incentivized and poorly compensated managers – while mostly true, is not actually helpful. Any useful advice must begin with theories about how to re-design a massively multisided platform to unhinge some sides and put new ones in.

The claim in this report is that any strategic re-engineering must begin by identifying the university's Hart asset (Davidson et al. 2020, Allen et al. 2022), which is that asset that cannot be externally owned without creating a hostage to the entire business. If the Hart asset is a credible warrant, then the institutional features that make a warrant credible are not costs but are properly accounted as the production technology of the asset. These include peer judgement rather than managerial direction, academic freedom, protected tenure for those who must be able to fail a candidate or reject a result, independence from the parties whose work is assessed, and corporate persistence beyond the tenure of any individual. The corollary is governance reforms that increase managerial control over academic judgement, and financial arrangements that make examiners dependent on the parties they examine, are strategically hazardous. They risk the institution's remaining strategic asset exactly when its scarcity value is rising. This does not require the whole institution to be governed collegially but only the certifying body.

6.4 Which university is this for?

The arguments above do not apply equally. Some universities are in a much less dangerous position *viz a viz* the AI disruption/annoyance/opportunity/apocalypse.

An elite university, whose asset is brand and reputation, is largely insulated: the mechanisms above leave it untouched, as it already sells the cohort and its real risks are complacency and the chance that a credential market in general disrepair drags its own signal down. A specialist university whose asset is deep expertise is exposed on content and protected on standing, since expertise tied to physical or regulated practice stays

scarce while expertise that is purely informational evaporates. Its strategy will always be speed into new domains and depth in the questions AI is pulling everyone else away from. It is the modal university, which is most universities (and whose Hart asset is administration) that these proposals are addressed to because they are the ones whose Hart asset has moved. In the long game, it may well come down to a choice between acquiring a new strategic asset and becoming a supplier of commodity instruction to somebody else's warrant. That's a great business opportunity for a school or research centre, but its market destruction for a university.

7. Conclusion

A thousand years ago, the first cohort of students sought out and hired the first suite of scholars to instruct and examine them. The scholars further thought to certify the result. The university was thus born as a platform and a warrant and remains so to this day. And everything since – the research university, mass participation, new media technologies, the multiversity, the rise of industry and alumni and parents' needs and governments' wants – has been elaboration on that transaction to add more sides to the increasingly complex object of governance. Now AI has arrived, bringing disruption.

The message of this report is that the disruption is real, and more dire than any previous technological shock to the university because it makes cheap the outputs by which knowledge has conventionally been verified. This is survivable, but requires structural change. The warrant remains valuable, indeed becomes more so as cheap output makes credible verification scarce. The cohort remains valuable and becomes more so as other forms of differentiation collapse. What does not survive is the assumption that these can be financed indefinitely by selling instruction at volume (instruction is now abundant). The romantic vision has no answer to this because it has no theory of revenue, and the industrial vision has no answer because a factory whose output has become cheap to counterfeit is not a factory with a quality problem but a factory in the wrong business. The practical route is greenfield. Reforming an incumbent multi-sided platform against its own participation constraints is slow but the technology is moving fast. New institutions can be designed around examination and warrant rather than instruction, adaptation rather than curriculum, and exploration rather than throughput. Existing universities can participate, but through separately governed and separately funded units, not through committee-led reform of the core. The university has survived a thousand years mostly by founding new colleges, not by rebuilding old ones.

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